Module III: Typesetting Mathematics

Study Notes · MAT5FS112 · Syllabus units 8–11

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1. Basic mathematical equations

1.1 Inline and displayed

Put maths between single `$` signs to keep it in the sentence: the area is A = πr^2. Use `\[ ... \]` (or the `equation*` environment) for a centred display:

```
$A = \pi r^2$

\[
  \int_0^1 x^2\,dx = \frac{1}{3}.
\]
```

$$\int_0^1 x^2\,dx = \frac{1}{3}.$$

1.2 Numbered equations and labels

The `equation` environment numbers the formula. A `\label` lets you refer to it with `\eqref`:

```
\begin{equation}
  E = mc^2 \label{eq:einstein}
\end{equation}
Equation~\eqref{eq:einstein} is the mass--energy relation.
```

$$E = mc^2 \tag{1}$$

Equation (1) is the mass–energy relation.

1.3 Scripts, fractions and roots

```
x^{2}          x_{i}          x_{i}^{2}       % superscript, subscript
\frac{a+b}{c}  \sqrt{x}       \sqrt[3]{x}     % fraction, roots
\binom{n}{k}                                  % binomial coefficient
```

Rendered: x^{2}, x_{i}, x_{i}^{2}, $(a+b)/c$, $\sqrt{x}$, $\sqrt[3]{x}$, $\binom{n}{k}$.

> **Braces group symbols.** Without them, only the next character is affected: `x^12` gives $x^{1}2$, while `x^{12}` gives x^{12}.

2. Groups of equations and numbering

`align` lines up several equations at the `&` character. Each row ends with `\\`. `align` numbers every row; the starred version `align*` numbers none.

```
\begin{align}
  (a+b)^2 &= a^2 + 2ab + b^2 \label{eq:sq} \\
  (a-b)^2 &= a^2 - 2ab + b^2 \nonumber
\end{align}
```

$$\begin{aligned} (a+b)^2 &= a^2 + 2ab + b^2 \qquad (2)\\ (a-b)^2 &= a^2 - 2ab + b^2 \end{aligned}$$

Row (2) is numbered and the second row is not, because `\nonumber` suppresses its number. A single row with a custom label uses `\tag{name}`:

```
\begin{equation}
  \sin^2\theta + \cos^2\theta = 1 \tag{$\star$}
\end{equation}
```

$$\sin^2\theta + \cos^2\theta = 1 \tag{★}$$

3. Matrices, dots, delimiters and affixed symbols

3.1 Matrices

Inside a display, the `pmatrix` (round brackets), `bmatrix` (square), `vmatrix` (determinant bars) and `Bmatrix` (curly braces) environments set matrices. Rows use `\\` and columns use `&`.

```
\[
  A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & 4 \\ 5 & 6 & 0 \end{pmatrix},
  \qquad
  \det\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.
\]
```

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & 4 \\ 5 & 6 & 0 \end{pmatrix} \qquad \det \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

3.2 Dots

```
\[
  x_1, x_2, \ldots, x_n \qquad
  a_{11} + \cdots + a_{nn} \qquad
  \vdots \quad \ddots
\]
```

Use `\ldots` on the baseline, `\cdots` for centred dots between operators, `\vdots` and `\ddots` for vertical and diagonal dots.

$$x_1, x_2, \ldots, x_n \qquad a_{11} + \cdots + a_{nn} \qquad \vdots \quad \ddots$$

3.3 Delimiters

Brackets that grow with their content use `\left` and `\right`:

```
\[
  \left( \frac{x}{y} \right)^{2} \qquad
  \left[ \sum_{k=1}^{n} k \right] \qquad
  \left\{ x \in \mathbb{R} : x > 0 \right\}
\]
```

$$\left(\frac{x}{y} \right)^{2} \qquad \left[\sum_{k=1}^{n} k \right] \qquad \left\{ x \in \mathbb{R} : x > 0 \right\}$$

Each `\left` needs a matching `\right`; use `\right.` when you want an invisible closing delimiter.

3.4 Affixed symbols (accents and decorations)

```
\[
  \hat{x} \quad \bar{x} \quad \vec{v} \quad \dot{x} \quad \ddot{x}
  \quad \tilde{a} \quad \overline{AB} \quad \underline{ab}
  \quad \overbrace{a+b+c}^{\text{sum}} \quad \underbrace{x+y}_{\text{pair}}
\]
```

$$\hat{x} \quad \bar{x} \quad \vec{v} \quad \dot{x} \quad \ddot{x} \quad \tilde{a} \quad \overline{AB} \quad \underline{ab}$$

4. Operators, relations and symbols

4.1 Greek letters

Lowercase letters are the name with a backslash: `\alpha`, `\beta`, `\gamma`, `\theta`, `\lambda`, `\pi`, `\sigma`, `\omega`. Capital letters are `\Gamma`, `\Delta`, `\Theta`, `\Lambda`, `\Sigma`, `\Omega`.

Rendered: $\alpha, \beta, \gamma, \theta, \lambda, \pi, \sigma, \omega$ and $\Gamma, \Delta, \Theta, \Omega$.

4.2 Big operators and limits

```
\[
  \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}, \qquad
  \prod_{i=1}^{n} i = n!, \qquad
  \lim_{x \to 0} \frac{\sin x}{x} = 1
\]
```

$$\sum_{n=1}^{\infty} 1/n^2 = \pi^2/6, \quad \prod_{i=1}^{n} i = n!, \quad \lim_{x \to 0} \sin x / x = 1$$

4.3 Named functions

Use `\sin`, `\cos`, `\log`, `\det` and `\max` rather than italic letters. For a name that LaTeX does not provide, use `\operatorname{name}` (from `amsmath`): `$\operatorname{sech}\,x = \dfrac{1}{\cosh x}$`.

4.4 Relations, arrows and set notation

```
\[
  x \le y, \quad x \ne y, \quad x \approx y, \quad
  f\colon A \to B, \quad x \mapsto x^2
\]
\[
  a \in A, \quad A \subseteq B, \quad A \cup B, \quad A \cap B, \quad
  \forall\, \varepsilon > 0\ \exists\, \delta > 0
\]
```

$$x \le y, \quad x \ne y, \quad x \approx y, \quad f\colon A \to B, \quad x \mapsto x^2$$

$$a \in A, \quad A \subseteq B, \quad A \cup B, \quad A \cap B, \quad \forall\, \varepsilon > 0\ \exists\, \delta > 0$$

4.5 Text and font inside maths

```
$\text{if } x > 0$        % text inside maths, uses the document font
$\mathrm{d}x$             % upright d for differentials
$\mathbf{v}$              % bold vector
$\mathbb{R}, \mathbb{N}$  % blackboard bold (amssymb)
$\mathcal{L}$             % calligraphic
```

Rendered: if $x > 0$, $\int f(x)\,\mathrm{d}x$ (upright *d*), $\mathbf{v}$, $\mathbb{R}$, $\mathbb{N}$, $\mathcal{L}$.

`\text{}` text inside maths	`\mathrm{}` upright letters	`\mathbf{}` bold	`\mathbb{}` blackboard bold
`\mathcal{}` calligraphic			

5. A worked example combining the tools

Let A be a 2×2 matrix with eigenvalues λ_1, λ_2. Then

```
\begin{align*}
  \operatorname{tr}(A) &= \sum_{i=1}^{2} \lambda_i, \\
  \det(A)              &= \prod_{i=1}^{2} \lambda_i .
\end{align*}
```

$$\operatorname{tr}(A) = \sum_{i=1}^{2} \lambda_i,$$

$$\det(A) = \prod_{i=1}^{2} \lambda_i.$$